

Mandatory Team Contribution Journal

Assignment A02: Comparative Analysis of ML/DL Frameworks

Houston City College – Fall 2026

Team Name / ID: DL_FA2026_2376_14562

Assigned Topic: TensorFlow vs. PyTorch

Primary Submitter: Michael Rivas

Individual Student Contributions

In accordance with course grading policies, team member responsibilities and verifiable contributions to Assignment A02 are detailed below:

1. Michael Rivas (Primary Submitter / System Administrator)

- Configured the Ubuntu Linux environment, managed dependency resolution (`pdflatex`, Python 3, Matplotlib, PyTorch, and TensorFlow libraries), and handled Git repository initialization and version control workflows.
- Authored the technical sections covering background history, JIT compilation mechanics (`torch.compile()` vs XLA), system integration, and real-world edge/cloud modularity.
- Assembled and formatted the final LaTeX report and compiled submission deliverables.

2. Cherluk (PyTorch Pipeline Architect)

- Spearheaded the design and execution of the PyTorch benchmarking pipeline.
- Implemented the Convolutional Neural Network (CNN) architecture on MNIST and integrated `torch.compile()` optimization loops.
- Collected empirical training duration and convergence telemetry across multiple experimental trials.

3. Olubukola (TensorFlow Infrastructure Lead)

- Developed the parallel TensorFlow model architecture utilizing Keras (`models.Sequential`) and managed data preprocessing pipelines.
- Configured and tested TensorFlow XLA JIT compilation (`@tf.function(jit_compile=True)`) to ensure structural and hyperparameter parity with PyTorch.
- Contributed comparative analysis regarding enterprise deployment and serving infrastructure.

4. Taki (Data Analysis & Telemetry Analyst)

- Conducted statistical variance and standard deviation analysis across all 10 experimental runs.
- Designed data visualization scripts and formatted empirical performance data into structured comparison tables.

- Verified cross-platform reproducibility and presentation timing.

Declaration: We hereby confirm that all listed members actively participated in the research, empirical benchmarking, and drafting of this assignment deliverables.

Signatures:

Michael Rivas: _____ Cherluk: _____

Olubukola: _____ Taki: _____